SAA6734AHL TFT-LCD panel controller

Highly-integrated SXGA flat-panel monitor controller with analog input and LVDS output

The SAA6734AHL controller for TFT panels integrates an analog VGA input (supporting up to 165 MHz RGB video), a scaler, provides an on-screen display (OSD) and has an LVDS output panel interface. It includes a host of on-chip features, lowering the cost and power consumption of next-generation SXGA (1280 x 1024 pixel) digital flat-panel monitors.

Semiconductors

Key features

- 165 MHz TFT-LCD controller
- Up to SXGA output resolution (1280 x 1024 pixels)
- Integrated ADCs (165 MHz for VGA interface)
- Philips' picture quality and contrast enhancements
- Advanced color processing for sRGB support
- Programmable horizontal/vertical up-scaling and down-scaling
- 10-bit gamma correction
- Robust OSD features include enhanced character OSD, rotation, flipping and more
- Standard I²C-bus interface
- Standard LQFP100 package for easy design in
- Complete reference design kit available.

IC family

Type no.	Output resolution	Max. input frequency
SAA6734AHL	SXGA	165 MHz
SAA6733AHL	XGA	110 MHz

Description

The feature-rich SAA6734AHL is a highly integrated TFT-LCD panel-controller IC that enables the manufacture of LCD monitors with a minimum number of components. An analog RGB VGA input is integrated on chip and supports up to 165 MHz video. Input resolutions up to UXGA (1600 x 1200 pixels) are allowed and inputs may be up- or down-scaled to fit panel resolutions of up to SXGA (1280 x 1024 pixels). The IC output comprises the popular LVDS interface standard.

The IC includes a versatile on-screen menu display (OSD) facility. This is a character-based OSD with predefined and programmable fonts. The IC includes an 8 kB font ROM, an 8 kB font RAM, and can display up to 1024 characters. Transparent or translucent overlay of OSD items is also possible and the character size, position and zoom factor are all fully programmable.

A low-cost version of the IC is available — the SAA6733AHL — which provides XGA output resolution (see table opposite). Both ICs come in the standard LQFP100 package, offer the analog VGA input, and are pin and register compatible.

The SAA6734AHL/33AHL controllers offer LCD monitor manufacturers outstanding video performance, design flexibility and cost savings. Moreover, a reference design kit is available to enable rapid product development.

Complete reference design kit

The reference design kit comprises a main board housing the SAA6734AHL IC, a keyboard and LCD panel connection accessories. All the necessary reference software is included and there is software library support for developing a full LCD monitor application.

PHILIPS

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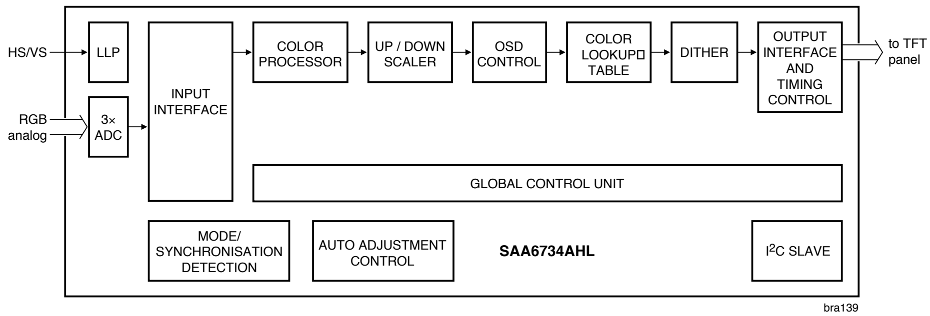

Input interface

- Analog input, triple ADC, 8 bit resolution, 165 MHz, standard and mid level clamping, automatic gain control
- Integrated PLL for dot clock recovery
- Maximum supported input resolution of 1600 x 1200 (UXGA)
- Freely definable data acquisition offsets window size (cropping)
- Composite sync, sync on green, tri-level sync on Y
- Integrated test-image generator
- Bi-directional monitor communication via DDC
- Auto detection:
  - Detection of presence and polarities of sync signals
  - Measurement of horizontal and vertical update frequencies
  - Statistical measurements to assist adjustment of sample clock frequency, vertical and horizontal sample offset and ADC parameters.

Image processing

- sRGB support
- Fully programmable color matrices supporting simultaneous color-space conversion and panel sRGB compensation
- Up-/Down-scaler with independent vertical/horizontal parameters
- Arbitrary scaling ratio from 0.25x to 64x
- Programmable vertical and horizontal polyphase filter supporting bicubic scaling with variable characteristics from smoothing to sharpening
- Temporal dithering of output to maintain precise 10-bit display on 8- and 6-bit panels
- Color correction look-up table (e.g. for gamma correction).

Output interface

- Display modes up to SXGA @ 85 Hz supported
- Single or dual pixel LVDS interface
- Generation of synchronization and validation signals for the Thin Film Transistor (TFT) display
- Freely programmable output timing supports displays from virtually any manufacturer
- Programmable output pin ordering.

Miscellaneous

- I^{2}C slave function for external microcontroller (4 selectable addresses)
- Cheap external crystal as master clock source and internal Phase-Locked Loop (PLL) panel clock generation from the system clock
- Unused function pins maybe used as GPIO
- Up to 3 PWM outputs
- Sophisticated power management.

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